PROJECT ABSTRACT

The **H-Bridge DC Motor Driver with PWM** project was developed as part of a workshop to understand the principles of DC motor control using power electronics and pulse-width modulation (PWM). The primary objective of this project was to design and implement a motor driver circuit capable of controlling the direction and speed of a DC motor efficiently. The project provided practical exposure to electronic circuit design, motor control techniques, and the application of switching devices in embedded systems.

An H-Bridge circuit was designed using MOSFETs to enable bidirectional rotation of the DC motor by reversing the polarity of the voltage applied to the motor terminals. PWM signals were used to regulate the average voltage supplied to the motor, allowing smooth speed control while maintaining high efficiency and reducing power loss. The circuit was first designed and verified through simulation before being assembled and tested on hardware. Various duty cycles of the PWM signal were applied to observe changes in motor speed, while forward and reverse operations were demonstrated by controlling the switching sequence of the H-Bridge.

The completed system successfully achieved reliable forward and reverse motor rotation along with variable speed control. The experimental results closely matched the simulation outcomes, confirming the effectiveness of the H-Bridge configuration and PWM technique. The project also enhanced the team's understanding of MOSFET switching characteristics, motor-driving principles, circuit protection, and troubleshooting methods.

This workshop project enabled the five-member team to gain hands-on experience in designing, simulating, assembling, and testing a practical motor control system. The knowledge gained through this project forms a strong foundation for advanced applications such as robotics, electric vehicles, industrial automation, conveyor systems, and embedded motor control. Overall, the project successfully demonstrated an efficient, cost-effective, and reliable method for controlling the speed and direction of a DC motor using an H-Bridge driver with PWM.